Embedded Continual Learning for Visual Identification on Resource-Constrained Automotive Hardware

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Project Status: In Progress

Primary Development Platform: NVIDIA Jetson Nano
Target Deployment Platform: Proprietary automotive vision-procession hardware

Overview. Modern automotive vision systems must identify objects and interpret visual information reliably while operating under strict constraints on computational resources, memory, power consumption and processing time. Conventional deep-learning systems can perform visual recognition effectively, but adapting them to new information may require retraining and substantial computational resources. Furthermore, updating a model can result in catastrophic forgetting, whereby previously acquired knowledge is lost.
This project investigates an embedded approach to visual identification that combines deep-learning feature extraction with vector-based comparison and continual learning. The objective is to explore how visual information can be identified and updated efficiently on resource-constrained hardware, while maintaining previously acquired knowledge.
Beyond the immediate machine-learning application, the project provides an opportunity to investigate the engineering challenges associated with deploying computationally intensive algorithms on resource-constrained hardware. These include processor utilisation, memory access, data movement, computational acceleration and hardware–software optimisation.
The work forms part of a research collaboration involving Lancaster University's Department of Engineering and School of Computing and Communications, alongside industrial research partners.

I. Aim and Objectives

Aim. To develop and evaluate an embedded visual-identification system and investigate the engineering considerations involved in implementing adaptive machine-learning algorithms on resource-constrained hardware.

The principal objectives are to:

1. Implement an image-processing pipeline on the NVIDIA Jetson Nano.
2. Investigate pretrained deep-learning models for extracting representative image feature vectors.
3. Implement vector-based assignment and similarity comparison for visual identification.
4. Evaluate the effects of model complexity, input resolution, batch size and reference dataset size.
5. Investigate continual-learning approaches that incorporate new information while preserving existing knowledge.
6. Measure computational performance, including latency, throughput and memory consumption.
7. Explore opportunities for hardware–software optimisation and deployment on specialised embedded processing architectures.

II. Technical Approach

The proposed system consists of three principal stages.
Image preprocessing and feature extraction: Input images are converted into a consistent RGB representation and processed by a pretrained deep-learning model. The model produces numerical feature vectors representing the visual characteristics of each image. Different model configurations and input resolutions will be investigated.
Vector-based identification: Extracted feature vectors are compared against stored reference vectors to identify previously encountered visual information. Candidate similarity measures include Euclidian distance and cosine similarity.
These approaches provide alternative methods of measuring similarity and will be evaluated in terms of identification performance and computational cost.
Continual learning: The system will be investigated for its ability to incorporate new visual information without requiring complete model retraining. Particular attention will be given to knowledge retention and catastrophic forgetting.

III. Experimental Methodology

Initial development and benchmarking will use the NVIDIA Jetson Nano. Experiments will examine how algorithmic choices influence both identification quality and hardware resource requirements.

Variable	Investigation
Model architecture	Full and lightweight feature-extraction models
Input resolution	Recognition performance versus computational cost

Reference dataset	Scaling from hundreds to thousands of images
Batch size	Comparison of batch sizes 8, 16 and 32
Similarity algorithm	Euclidean distance and cosine similarity
Continual learning	Adaptation and retention of previously acquired information

Performance will be evaluated using identification accuracy, processing latency, throughput, memory consumption and scalability. Where possible, feature-extraction and vector-comparison costs will be measured separately to identify computational bottlenecks.

IV. Broader Engineering Applications

The computational requirements of embedded machine learning create several engineering problems that are closely connected to embedded systems architecture and hardware optimisation.

Hardware–software co-design: Different machine-learning algorithms impose different demands on processors, memory systems and specialised accelerators. Understanding these requirements can inform the selection and configuration of embedded hardware, as well as the design of software pipelines that make efficient use of available processing resources.

Hardware acceleration: Deep-learning inference and vector comparison involve substantial numerical computation. Their underlying operations provide opportunities for parallel processing and acceleration using GPUs, FPGAs or application-specific hardware. Investigating which operations dominate execution time can help identify suitable candidates for hardware acceleration and guide the partitioning of workloads between processing units.

Memory architecture and data movement: Feature extraction and reference-vector comparison can involve repeated access to large datasets. Memory capacity, bandwidth, access patterns and data movement may therefore become significant performance constraints. Efficient data organisation, caching, memory reuse and reduced intermediate storage are potential approaches to improving system performance.

Real-time embedded processing: Automotive and other safety-critical embedded applications require predictable processing behaviour. Beyond average inference speed, engineering considerations include worst-case latency, processing deadlines, resource contention and reliable operation under varying workloads.

Power and computational efficiency: Embedded platforms often operate within strict energy and thermal budgets. Model compression, reduced numerical precision, efficient memory utilisation and workload scheduling provide potential methods of reducing computational cost while maintaining acceptable identification performance.

Edge intelligence and automotive systems: The same principles can be applied to intelligent sensing, driver-assistance systems, object detection, road-scene interpretation, anomaly detection and other applications requiring local processing. These applications demonstrate the broader relevance of combining adaptive algorithms with efficient embedded hardware.

V. Expected Outcomes

The project is expected to produce:

- An embedded image-processing and deep-learning feature-extraction pipeline.
- A vector-based visual-identification implementation.
- Experimental evaluation of identification performance and computational requirements.
- An investigation of continual learning and knowledge retention.
- Identification of computational and memory bottlenecks in embedded visual processing.
- An engineering assessment of opportunities for hardware acceleration, memory optimisation and hardware–software co-design.
- A foundation for investigating deployment on specialised automotive processing architectures.

The overall objective is to demonstrate how adaptive visual-identification algorithms can be evaluated not only as machine-learning models, but also as computational workloads within real embedded systems. This provides a basis for further research into efficient, reliable and application-specific hardware implementations.